

MS-BASIC worked with LOAD and SAVE commands.

 flexonsbd.blogspot.com/2018/08/ms-basicload-save.html

Thursday, August 2, 2018

MS-BASIC ran with LOAD and SAVE commands.

I tried running Microsoft BASIC on a 6809SBD.

MS-BASIC is running on [Vintagechips ' SBC6809. \(The port was done by tomi9jp \)](#)

On my main environment, Flex9, I have the BASIC compiler from TSC, so it's not particularly necessary, but since it's MS-BASIC that I used back in the PC-8801 days, I ported it to Flex9 out of nostalgia. First,

download BASIC9.zip from [Vintagechips](#) ' OneDrive, change the ACIA address in BASIC9.asm to \$F0BC,BD, and also change the start address to \$A000 and the end of memory to \$9FFF, then put it into FlexDrvWin and assemble it with Flex9's assembler ASMB. A lot of errors will occur due to the difference in assembler format, so you have to correct them manually.

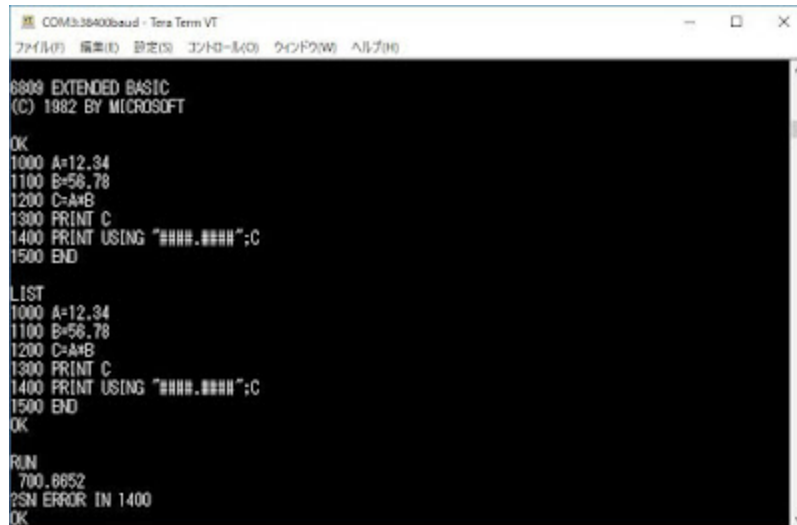
For example CMPA #'E' -> CMPA #E

Also, Flex9's ASMB has a problem where it only distinguishes labels up to 6 characters long, so you get a confusing error where, for example, TOK_INPUT and TOK_INKEY are the same label.

I renamed the three conflicting labels.

For example, TOK_INPUT was in conflict with TOK_INKEY, so I changed it to TOK_IK.

There were a few other minor modifications, but I was able to assemble it successfully and create BASIC9.CMD. When I ran it, I saw the familiar Microsoft BASIC message...



```
COM3:38400baud - Tera Term VT
ファイル(F) 編集(E) 設定(S) コントロール(C) ウィンドウ(W) ヘルプ(H)

0800 EXTENDED BASIC
(C) 1982 BY MICROSOFT

OK
1000 A=12.34
1100 B=56.78
1200 C=A*B
1300 PRINT C
1400 PRINT USING "####.####";C
1500 END

LIST
1000 A=12.34
1100 B=56.78
1200 C=A*B
1300 PRINT C
1400 PRINT USING "####.####";C
1500 END
OK

RUN
 700.8652
25N ERROR IN 1400
OK
```

BASIC9

However, PRINT USING doesn't work!!! (Perhaps it was removed due to ROM size limitations?)

Also, it's frustrating that I can't save or load the programs I've created!

So, I ported the ExBasRom from Grant's website, which is the original source, and added SAVE and LOAD commands.

Ported ExBasRom and added SAVE and LOAD functions.

First, download ExBasRom.zip from [Grant's website](#) and extract ExBasRom.asm from it. Change the ACIA address to \$F0BC,BD, and since SAVE and LOAD routines will be added, change the start address to \$9000 and the end of memory to \$8FFF. Make other modifications in the same way as above.

Next, in order to add the SAVE and LOAD routines, add the following four lines FCC "SAV"
9F FCB \$80+'E' FCC "LOA" A0 FCB \$80+'D' after

FCC "EDI " 9E FCB \$80+

'T around line 590. As a result, the lines from "TAB" to TOK_USING will shift, so change about 12 places to values that are +2, such as \$9F->\$A1, ... \$B1->\$B3. Furthermore, insert the following two lines, FDB SAVE 9F and FDB LOAD A0, between FDB EDIT 9E and TOK_HIGH_EXEC EQU ... around line 745. Finally, place the main body of the SAVE and

LOAD routines after *LINE LINE CMPA ... LBEQ ... JMP LB277 around line 5400.

Regarding the scope of SAVE, the program body has the starting address in the work area TXTTAB(\$0019,1A) and the last address + 1 in VARTAB(\$001B,1C), but to save values in other work areas as well, it saves from address \$0000 to the value of VARTAB - 1. Put the modified ExBasRom.asm into FlexDrvWin and assemble it with the Flex9 assembler ASMB.

With this, we have completed an MS-BASIC script where PRINT USING works and the SAVE and LOAD commands are available.

The size was \$9000 to \$B595, totaling \$2596 (9622) bytes.

```
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6800 EXTENDED BASIC
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OK
1000 A=12.34
1200 B=56.78
1300 C=A*B
1400 PRINT C
1500 PRINT USING "####.####":C
1600 END

LIST
1000 A=12.34
1200 B=56.78
1300 C=A*B
1400 PRINT C
1500 PRINT USING "####.####":C
1600 END
OK
RUN
700.6652
700.6652
OK
```

ExBasRom

The SAVE and LOAD routines are also working correctly. If no drive is specified, the save location is the Flex9 work drive.

```
COM3:38400baud - Tera Term VT
ファイル(F) 編集(E) 設定(S) コントロール(C) ウィンドウ(W) ヘルプ(H)

LIST
1000 A=12.34
1200 B=56.78
1300 C=A*B
1400 PRINT C
1500 PRINT USING "####.####":C
1600 END
OK
RUN
700.6652
700.6652
OK
SAVE
SAVE FILE NAME : TESTPRG
OK
NEW
OK
LIST
OK
LOAD
LOAD FILE NAME : TESTPRG
OK
LIST
1000 A=12.34
1200 B=56.78
1300 C=A*B
1400 PRINT C
1500 PRINT USING "####.####":C
1600 END
OK
RUN
700.6652
700.6652
OK
```

ExBasRom with SAVE and LOAD functions

Vintagechips and tomi9jp's porting work was extremely helpful in this process. Without their efforts and achievements, it would have taken much more time and effort. Thank you very much. I've uploaded

the source code ExBasROM_KAI.TXT, which has been modified for Flex9 and includes PRINT USING, SAVE, and LOAD functions, to [Microsoft's OneDrive](#).